

Product Deep Dive

Product Needs

- Mobile-Optimized Custom Mezo web page for card activation and mezo id reservation
- Web page / interface on a laptop with RFID reader
- New DB table for managing card data
- Potentially a new onboarding feature for “claiming a reserved Mezo ID” (see “tricky part”)

User Experience Thoughts

Happy path makes sense to me, but there's a few things we'd have to think about.

So the user's card would have two pieces of data on it: the unique RFID UID (unique to the card) and a QR code. [I'm guessing this is not unique and would simply lead to mezo.org/vegasbaby](#). (yes that should be the URL).

- I was planning on the QR codes being unique. either we have to stick them on ourselves (not happy path lol, but enough team and enough beers - ez task) Or the fulfillment company can add individual QRs (unlikely now that you bring it up) So the QR was going to be the unique link tied to the RFID UID in the DB

Now the user scans the non-unique QR code and is sent to a sharp mezo landing page that can do two things:

1. Let's the user enter their email (verifies correct format, all good)
2. Let's the user enter a preferred Mezo ID for “reservation” (needs to verify availability)

Email is easy & for the Mezo ID we'd require a check there to hit our DB wherever Mezo IDs are and make sure the ID is not taken, and error accordingly until a unique name was entered. We'd then need to understand how to “mark the name as reserved” ([this ticket](#) could give us some insight there as it's something we need to do).

Linking this back to a true “mezo account” is the tricky part. I'll skip for now, but go to [Tricky Part](#) for more.

User confirms email and an available name to reserve. Hits enter. Sees confirmation. [Maybe get's confirmation email?](#)

Continuing with the experience, we now assumedly have this new vegas DB/table and it has (card_uid, activated, balance, last_active). Roughly imagining this data as:

- card_uid: 04:A2:BB:1C:99 (usually 4-7 bytes, w/e the manufacturer uses)
- activated: 1 or 0
- balance: 5 (if we're using 5 as amount, we'd expect this to only ever be 5, 4, 3, 2, 1 or 0)
- last_spend: 2025-04-18 14:52:00 (date of last scan/deduction)

User visits the Mezo booth and their card is scanned by an RFID reader. This application takes their card_uid and checks against the new db/table to display activation and balance. This application also allows the booth user to “debit” balance in “balance” column. So think a button for “Coffee: \$1” and another for “Hoodie \$5”. Once these buttons are clicked, the DB is updated to reflect the appropriate amount. At some point when full spend is reached this application should read “Balance 0”. Application also has to handle edge cases where a debit is requested for \$3 on a UID that only has \$1 (e.g. not enough “funds”).

Which reminds me: [I'm assuming this is all “funny money” / not actually MUSD / “just numbers in a spreadsheet” which I think is the case here.](#) (This is the case)

Fallback Issue

Note that if the RFID reader is not working you have no way to input a card_uid if that is indeed the key value in the table. If the RFID cards come with the number printed on them (a unique ID for each card, the ID burned into the RFID chip) then you could

get around this with an input field.

Tricky Part

Okay, now for the tricky part. If you implement the flow as described above (seems realistic bc it's hacky), you've got this issue of having a Mezo ID reserved in our database without a supabase account set up. I have to double check this with my team (Kuba) though I'm fairly certain we require a wallet connection and signature to create what essentially is the mezo account (e.g. it's actually a supabase UUID in our database that has a paired wallet address). This action happens when a user signs with their wallet during onboarding. Onboarding also requires the terms and conditions check in addition to the requirement of having a web3 wallet to sign in order to create this account. It's also where the mezo id reservation step is atm.

Since we'd want to push a mobile landing page, passport integration would not work here (similar if you head to mezo.org or pamp.land atm, we do not have mobile support). Also, the account creation/onboarding flow is unique to Cathedral right now and is not on the passport npm library (e.g. no current way to "onboard" with wallet + t&c + select mezo username off cathedral atm). I'm also thinking this might pose some challenges as we have not stress tested. Thirdly, I'm assuming we didn't want to gate it by having to have the user sign up with a wallet (which would require desktop) as it would impact the conversion in a non significant way I'd wager.

So the problem we need to solve is if we do go the scrappy route, we'd need a way for when that user comes back and logs in, we have a mechanism for creating an account (onboarding) where email is added, and then we double check the new table to see if a mezo id was reserved and then apply that if they log in with their wallet. This seems possible, but would be a good amount of work tbh.

Other Notes

An RFID reader like this hooked up to a laptop should do the trick just fine fyi:

https://www.amazon.com/ACR122U-Reader-Writer-Ntag213-Software/dp/B07KRKPWYC?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&psc=1&smid=A3K2PHG2D8M445&gQT=1